

Midterm 2 Review Session

Physics 7B — Electric field, Gauss's law, potential, and capacitors

Contents

1	Electric Charge and Electric Field	2
2	Gauss's Law	5
3	Electric Potential	7
4	Capacitors	9
5	Energy and Motion in Electric Fields	13

Electric Charge and Electric Field

Mini-Lecture

Coulomb's law gives the force between two charges Q_1, Q_2 ,

$$F = k \frac{Q_1 Q_2}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{Q_1 Q_2}{r^2},$$

where r is the distance between the two charges. In components, you must use the full geometric distance r first and only then resolve the result into x - and y -components.

The electric field is defined by $\mathbf{E} = \mathbf{F}/q$ for a test charge q . So the electric field due to a point charge Q is

$$E = \frac{F}{q} = k \frac{Q}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}$$

By extension, the force exerted on a charge q placed in an electric field is given by $\mathbf{F} = q\mathbf{E}$.

A few useful reminders:

- Positive charges are sources for field lines and negative charges are sinks.
- Positive charges move with the field and negative charges move against it.
- Electric fields obey superposition: if one source gives $\mathbf{E}_1$ and another gives $\mathbf{E}_2$, then the total field is $\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2$.

For a continuous charge distribution, replace Q with dQ and integrate:

$$dE = \frac{1}{4\pi\epsilon_0} \frac{dQ}{r^2}$$

The charge element depends on the geometry:

- **Line charge:** $dQ = \lambda d\ell$
- **Surface charge:** $dQ = \sigma dA$
- **Volume charge:** $dQ = \rho dV$

In practice, the hard part is usually writing the denominator r^2 and the differential element in terms of your integration variables.

The electric field in a perfect conductor is zero, and all excess charge on a perfect conductor lives on its surface.

Finally, for an electric dipole with dipole moment $\mathbf{p}$, the torque and potential energy are

$$\tau = \mathbf{p} \times \mathbf{E} \quad U = -\mathbf{p} \cdot \mathbf{E}$$

Example Problem: Why not just make it a semi-circle?

Suppose you have a uniformly charged unclosed circle of charge Q , radius 1, centered at $(0,0)$, such that there is no circle above $y = 1/\sqrt{2}$. Find the electric field at the origin.

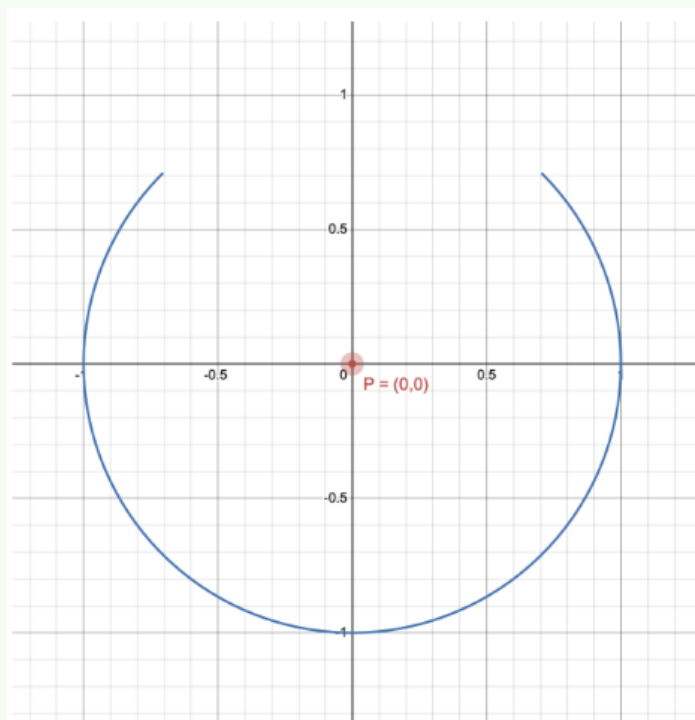


Figure 1: Electric Field Problem Figure

Example Problem: Bob's issue with E&M

Bob is a positive guy, so positive he flows with charge $+Q$. He's friends with a downer, named Paul, who's so negative he flows with charge $-Q$. One day, Bob is hanging out with Paul and they realize that there's an external electric field being applied to them, with

$$\mathbf{E}_{\text{ext}} = E \hat{x}, \quad E > 0$$

Like a beaver with running water, Bob simply can't have this happening at the point between him and Paul, so he wants to position himself relative to Paul on the origin such that at the point between him and Paul the net electric field is 0.

- (a) Where should Bob position himself?
- (b) Suppose the external electric field suddenly shifts such that it is pointing in the positive $\hat{y}$ direction. How much work does Bob have to do to cancel this new electric field?

Practice Problem: Charged tube on its axis

An uncapped tube of radius R and length L carries uniform surface charge density σ .

- (a) Explain why you cannot use Gauss' law to determine the electric field created by the cylinder at any point on the x -axis.
- (b) Determine the electric field created by a very thin slice of the uniformly charged tube at any point on the x -axis. Take the origin at the center of the thin ring.
- (c) Determine the electric field created by the charged tube at any point on the x -axis. Take the origin at the right end of the cylinder.

Problem 2 - Charged tube (20 pts)

An uncapped tube of radius R and length L (Fig.2) carries uniform surface charge density σ .

- a) Explain why you cannot use Gauss' law to determine the electric field created by the cylinder at any point on the x -axis.
- b) Determine the electric field created by a very thin slice of the uniformly charged tube at any point on the x -axis. Take the origin at the center of the thin ring.
- c) Determine the electric field created by the charged tube at any point on the x -axis (Fig.2). Take the origin at the right end of the cylinder.

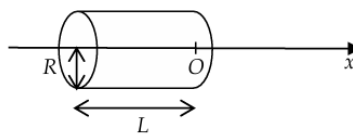


Figure 2

Gauss's Law

Mini-Lecture

The electric flux through a surface is given by

$$\Phi_E = \mathbf{E} \cdot \mathbf{A} = EA \cos \theta,$$

where $\mathbf{A}$ points perpendicular to the surface. In integral form,

$$\Phi_E = \int \mathbf{E} \cdot d\mathbf{A}$$

Gauss's law is

$$\oint \mathbf{E} \cdot d\mathbf{A} = \frac{Q_{\text{encl}}}{\epsilon_0}$$

The law is always true, but it becomes directly useful only when symmetry lets us choose a Gaussian surface on which the electric field is either constant and perpendicular to the surface, or parallel to it so that the flux vanishes.

The three standard geometries are:

- **Sphere:** with spherical symmetry,

$$4\pi r^2 E = \frac{Q_{\text{encl}}}{\epsilon_0} \quad \Rightarrow \quad E = \frac{Q_{\text{encl}}}{4\pi\epsilon_0 r^2}$$

- **Infinite cylinder / line:** with cylindrical symmetry,

$$2\pi s L E = \frac{Q_{\text{encl}}}{\epsilon_0} \quad \Rightarrow \quad E = \frac{Q_{\text{encl}}}{2\pi\epsilon_0 s L}$$

If $Q_{\text{encl}} = \lambda L$, then

$$E = \frac{\lambda}{2\pi\epsilon_0 s}.$$

- **Infinite sheet / slab:** using a pillbox,

$$2AE = \frac{Q_{\text{encl}}}{\epsilon_0} \quad \Rightarrow \quad E = \frac{Q_{\text{encl}}}{2A\epsilon_0}$$

If $Q_{\text{encl}} = \sigma A$, then

$$E = \frac{\sigma}{2\epsilon_0}$$

If the enclosed charge is not given directly, compute it from the appropriate density:

$$Q = \int \lambda d\ell \quad Q = \int \sigma dA \quad Q = \int \rho dV$$

Example Problem: Gauss - shmauss, as long as I'm right

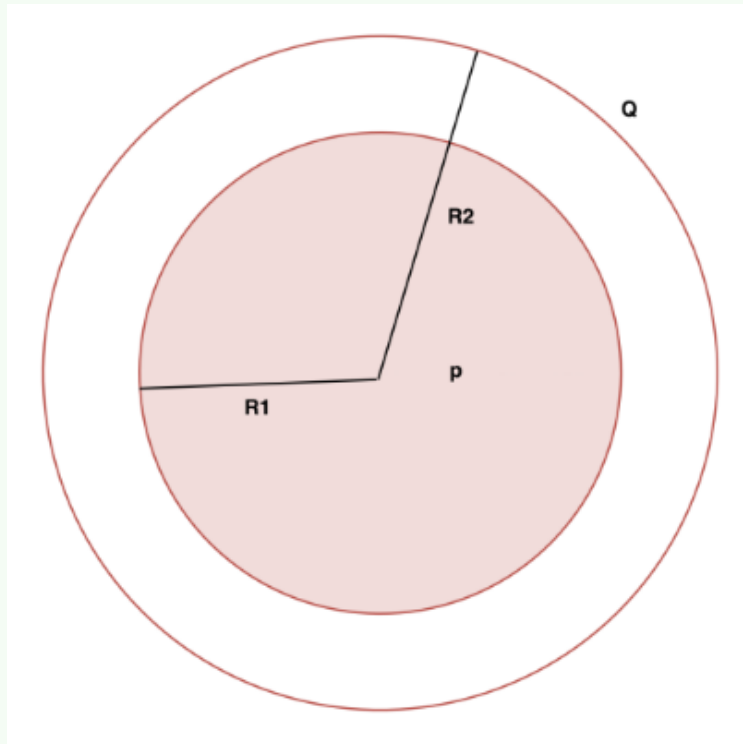
Paul makes a bet with Bob that he can easily deduce what the electric field is outside of an infinitely long cylinder with radius R_1 and volume charge density ρ that Bob gets to choose, as long as he is able to place an infinitely thin but infinitely long cylinder of radius R_2 on the outside of the cylinder with a charge Q that he can choose.

- (a) Bob decides to make the inner cylinder have a volume charge density of

$$\rho(r) = \frac{Q_0 r^2}{2\pi}.$$

What is the electric field for $r < R_2$?

- (b) Paul says, "Great job Bob, I'll have the outer cylinder have the negative of whatever total charge you placed on the inside. The outside E -field is just 0." What charge per unit length is needed on the outer ring?



Electric Potential

Mini-Lecture

The electric potential is a scalar function satisfying

$$\mathbf{E} = -\nabla V$$

That means the potential difference between two points is

$$V(b) - V(a) = - \int_a^b \mathbf{E} \cdot d\ell$$

If you use infinity as the reference point, then it is safest to write

$$V(\infty) - V(a) = - \int_a^\infty \mathbf{E} \cdot d\ell$$

Usually we choose $V(\infty) = 0$.

For a point charge,

$$V = \frac{1}{4\pi\epsilon_0} \frac{Q}{r}$$

For a continuous charge distribution, use

$$dV = \frac{1}{4\pi\epsilon_0} \frac{dQ}{r}$$

and integrate. Since V is a scalar, it is often easier to compute the potential first and then take a derivative to recover the electric field if needed.

Conductors are equipotentials. The potential energy of a charge q in a potential V is

$$U = qV$$

For a dipole,

$$V = \frac{1}{4\pi\epsilon_0} \frac{p \cos \theta}{r^2}$$

Example Problem: A potentially fun problem?

- What is the potential at the origin of Problem 1.1 if we say that the potential at $y = \infty$ is $V(\infty) = 0$? What is the speed of an electron, with charge $-e$ and mass m_e , at the origin if it is released from rest at $y = \infty$?
- If we neglect the outer cylinder in Problem 2.1, then what is the potential outside some distance r from the center of the inner cylinder? Define 0 potential to be at distance R_2 . Qualitatively, what would the potential be in the cylinder if the inner cylinder was instead a conductor?

Practice Problem: CO₂ molecule

The CO₂ molecule can be represented by a linear triplet of point charges

$$\{A(-q), O(+2q), B(-q)\}$$

such that $AO = OB = a$, as shown.

- Determine the electric potential $V(r, \theta)$ at any point M such that $r \gg a$.
- Explain how you would calculate the components (E_r, E_θ) of the electric field created by this charge distribution. You do not need to calculate them.

Hints: distance r_A can be written as

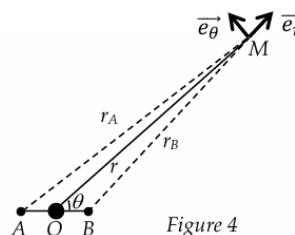
$$r_A = \left((\vec{AO} + \vec{OM})^2 \right)^{1/2},$$

and additionally you may expand

$$(1 + \varepsilon)^{-1/2} \approx 1 - \frac{\varepsilon}{2} + \frac{3\varepsilon^2}{8} \quad (\varepsilon \ll 1).$$

Problem 4 - CO₂ molecule (20 pts)

The CO₂ molecule can be represented by a linear triplet of point charges: $\{A(-q), O(+2q), B(-q)\}$ such that $AO=OB=a$, as shown in Fig.4.



- Determine the electric potential $V(r, \theta)$ at any point M such that $r \gg a$.

Hints: Distance r_A can be written as $\left((\vec{AO} + \vec{OM})^2 \right)^{1/2} = (\vec{AO}^2 + 2\vec{AO} \cdot \vec{OM} + \vec{OM}^2)^{1/2}$. Additionally, the function $(1 + \varepsilon)^{-1/2}$ should be expanded to the second order in ε , where $\varepsilon \ll 1$.

$$(1 + \varepsilon)^{-1/2} \approx 1 - \frac{\varepsilon}{2} + \frac{\varepsilon^2}{8}$$

- Explain how you would calculate the components (E_r, E_θ) of the electric field created by this charge distribution. You do not need to calculate them.

Hint: The polar basis $(\vec{e}_r, \vec{e}_\theta)$ -which is a truncated cylindrical basis- represented at point M might be useful.

Capacitors

Mini-Lecture

A capacitor is made of two conductors. If charges $+Q$ and $-Q$ are placed on the conductors, then there is a potential difference V between them, and the capacitance is defined by

$$C = \frac{Q}{V}$$

To find the capacitance of a pair of conductors:

1. Put charges $+Q$ and $-Q$ on the conductors.
2. Find the electric field between them.
3. Integrate the field from one conductor to the other to get the potential difference V .
4. Compute the positive quantity $C = |Q/V|$.

Useful standard results:

$$C_{\text{parallel plate}} = \frac{\epsilon_0 A}{d} \quad C_{\text{cyl shell}} = \frac{2\pi\epsilon_0 L}{\ln(b/a)} \quad C_{\text{spherical shell}} = \frac{4\pi\epsilon_0}{\frac{1}{a} - \frac{1}{b}}$$

Capacitors store energy,

$$U = \frac{1}{2} \frac{Q^2}{C} = \frac{1}{2} QV = \frac{1}{2} CV^2$$

If the space between the conductors is filled with a dielectric of constant K , then the capacitance becomes

$$C = KC_0$$

For circuits:

- **Parallel:** same voltage, so $C_{\text{eq}} = C_1 + C_2 + \cdots + C_n$.
- **Series:** same charge, so

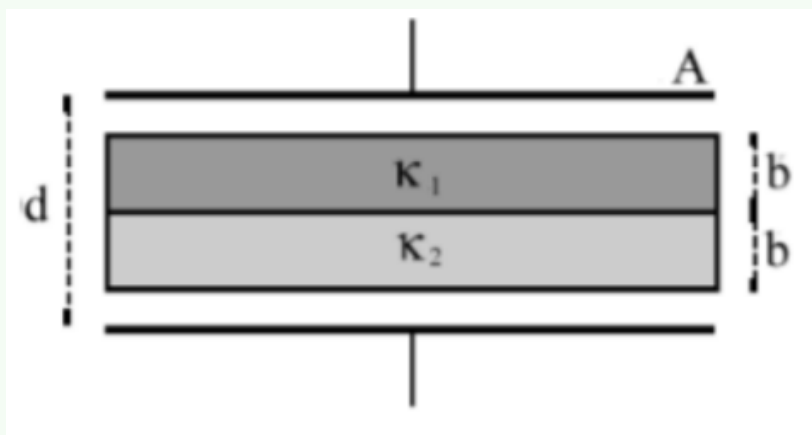
$$\frac{1}{C_{\text{eq}}} = \frac{1}{C_1} + \frac{1}{C_2} + \cdots + \frac{1}{C_n}$$

For partially filled capacitors, it is often helpful to reinterpret the geometry as several simpler capacitors connected in series or in parallel.

Example Problem: A Capacity to Learn

A parallel-plate capacitor, with area A and distance d between the plates, is filled with two different dielectrics as shown. Assume the top plate is negatively charged and $\kappa_1 > \kappa_2$.

- Assume that the plates of the capacitor carry charge $\pm Q$. Sketch the electric field as a function of distance from the negative plate.
- Sketch the potential $V(x)$ within the capacitor, where x is the distance from the negative plate. What is the total potential difference across the capacitor?
- Find the capacitance of this configuration.

Example Problem:**Example Problem: A Capacity to Learn More**

A parallel-plate capacitor, with area A and distance d between the plates, is filled with two different dielectrics as shown. Each dielectric covers half the surface area of the plates.

- Find the capacitance. (Hint: Regard this as two capacitors in parallel.)
- If the charge on the plates is $\pm q$, then find the electric field in the two regions.

Practice Problem: Coaxial capacitor & dielectric

Two coaxial cylindrical conducting shells of identical length L and respective radii R_1 and R_2 ($R_2 > R_1$) carry uniformly distributed electric charge $+Q$ and $-Q$ respectively. You may assume that $L \gg R_2$.

- Calculate the electric field at any point located a radial distance r from the symmetry axis.
- Calculate the absolute value of the voltage between the two conducting shells.
- Determine the capacitance of this cylindrical capacitor.
- Calculate the capacitance if the gap between the shells is filled by two successive dielectric materials of dielectric constant K_1 and K_2 , with inner dielectric thickness d as shown.

Problem 3- Capacitor & dielectric (20 pts)

Two coaxial cylindrical conducting shells of identical length L and respective radii R_1 and R_2 ($R_2 > R_1$) carry uniformly distributed electric charge $+Q$ and $-Q$ respectively ($Q > 0$). They are separated by a vacuum of permittivity ϵ_0 . You may assume that $L \gg R_2$. A cross-section view is presented in Figure 3a.

- Calculate the electric field created at any point M located at a radial distance r from the symmetry axis. *Show your work!*
- Calculate the absolute value of the voltage between the 2 conducting shells.
- Determine the capacitance of this cylindrical capacitor.
- Calculate the capacitance if the gap between the 2 shells is filled by 2 successive dielectric materials of dielectric constant K_1 and thickness d for the inner one, and dielectric constant K_2 for the outer one, as sketched in Figure 3b.

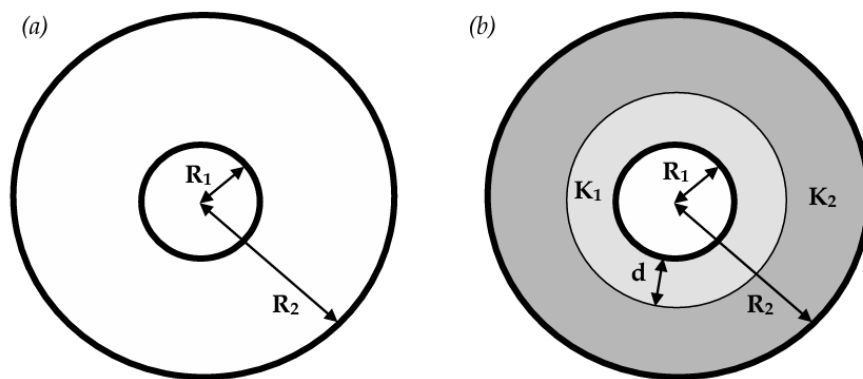


Figure 3: cross-section views

Practice Problem: Coaxial cylindrical shells in oil

Two metallic coaxial cylindrical shells are placed vertically in a bucket containing oil of density ρ and dielectric constant κ . The length L of the shells is assumed to be very large compared to the inner radius a and outer radius b . A constant voltage V is applied across the two conducting shells, and the oil rises to a height h between the shells.

- Determine the capacitance when there is no oil between the shells.
- Determine the capacitance when the oil reaches height h between the shells.
- Calculate ΔU_{batt} and ΔU_{capa} , the changes in potential energy of the battery and capacitor resulting from the change in the height of the oil from 0 to h .
- Determine the sign of the change in the total potential energy of the battery and capacitor, and explain physically where the energy comes from or goes.

Problem 5 - Capacitor (20 pts)

Two metallic coaxial cylindrical shells are placed vertically in a plastic bucket containing some oil of density ρ and dielectric constant κ . The length L of the shells is assumed to be very large compared to the inner radius a and outer radius b . A constant voltage V is applied across the two conducting shells, and the oil is observed to go up to a height h between the plates, as shown in Fig.5.

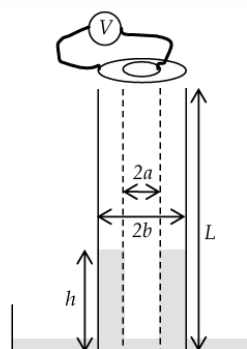


Figure 5

- Determine the capacitance of the cylindrical capacitor when there is no oil between the shells.
- Determine the capacitance of the cylindrical capacitor when the oil reaches height h between the shells.
- Calculate ΔU_{batt} and ΔU_{capa} , the changes in potential energy of the battery and of the capacitor, resulting from the change in the height (from 0 to h) reached by the oil between the shells.
Hint: ΔU_{batt} is the negative of the work done by the battery to push some charge ΔQ on the capacitor's plates.
- Determine the sign of the change in total potential energy of the battery and capacitor. If $\Delta U_{\text{tot}} > 0$, explain where the added energy comes from, and if $\Delta U_{\text{tot}} < 0$, explain where the released energy is stored.

Energy and Motion in Electric Fields

Mini-Lecture

When a particle moves in an electrostatic field, energy methods are often the fastest route. The main relations are

$$\Delta U = q \Delta V \quad K_i + U_i = K_f + U_f$$

If the particle starts from rest, then

$$K_f = U_i - U_f = q(V_i - V_f)$$

This is especially useful when the electric potential is easier to compute than the full vector electric field.

Practice Problem: Electron released along the axis of a charged disk

An electron of mass m and electric charge $-e$, initially at rest, is released from infinity along the symmetry axis of a uniformly charged disk of radius R . The disk carries positive surface charge density σ . Calculate the kinetic energy of the electron at distance z from the center of the disk.

Problem 4 - Electric potential & potential energy (20 pts)

An electron of mass m and electric charge $-e$, initially at rest, is released from infinity along the symmetry axis of a uniformly charged disk of radius R . The flat disk carries positive surface charge distribution σ . Calculate the kinetic energy of the electron at distance z from the center of the disk (Fig.4). You may assume that the gravitational potential energy is negligible.

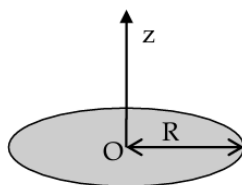


Figure 4

Practice Problem: Electric field & potential from a model atom

We model an atom by a spherical negative charge distribution $\rho(r)$ extending beyond the radial distance $a > 0$ around a nucleus of charge $q > 0$. The electric potential is given by

$$V(r) = \frac{q}{4\pi\epsilon_0 r} e^{-r/a}$$

- (a) Determine the direction and magnitude of the electric field $\mathbf{E}(r)$.
- (b) Calculate the electric flux $\Phi_E(r)$ through a sphere of radius r .
- (c) Calculate the total enclosed charge $Q_{\text{tot}}(r)$. What is the limit as $r \rightarrow \infty$?
- (d) Determine the negative charge distribution $\rho(r)$.

Problem 5 - Electric field & potential (25 pts)

We assume that atoms can be modeled by considering a spherical negative charge distribution $\rho(r)$ extending beyond the radial distance a ($a > 0$), around the nucleus of charge q ($q > 0$).

At distance r from the center O of the atom, the electric potential is given by the following expression:

$$V(r) = \frac{q}{4\pi\epsilon_0 r} e^{-r/a}$$

- a) Determine the direction and magnitude of the electric field $\vec{E}(r)$ created by this charge distribution at a distance r from the origin.
- b) Calculate the flux $\Phi_E(r)$ of the electric field through a sphere of center O and radius r .
- c) Calculate the electric charge Q_{tot} enclosed in the sphere of center O and radius r . What is the limit of Q_{tot} when $r \rightarrow \infty$? Interpret your result.
- d) Determine the negative charge distribution $\rho(r)$.
Hint: it might be helpful to express the infinitesimal amount of negative charge dQ contained in a spherical shell of thickness dr .